Short Simulators, Long Experiments: A Controlled Audit of Horizon Mismatch in Simulation-Based Inference for Live-Cell Microscopy

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Abstract

Amortized simulation-based inference (SBI) may be trained on short synthetic recordings and then applied to consecutive windows of longer experiments. Unlike independent training clips, later windows inherit molecular composition, spatial correlations, and photophysical history. We quantify this horizon mismatch in a reaction–diffusion model of receptor dimerization with synthetic live-cell microscopy. For each of 500 parameter settings, we compare ten independent 2 s simulations with ten consecutive 2 s windows from one 20 s simulation, using exact simulator truth. We examine the dimer-complex fraction, dimer unbinding rate, and monomer diffusion coefficient. By the final window, errors in the first two exceed prespecified tolerances, whereas diffusion remains within tolerance. For dimer composition, 90% interval coverage falls from 0.95 to 0.39 without wider posterior intervals. Averaging windows reduces total error but not the history-induced error. We use its simulated magnitude as a sensitivity test, not a correction, for experimental MET recordings. The difference between resting and activated conditions remains much larger than this effect. However, absolute fractions remain model-dependent, and the workflow does not support interpreting a whole-recording unbinding estimate as a single constant rate. Short-window SBI should therefore be validated separately for each inferred quantity and aggregation rule.

1 Introduction

Simulation-based inference (SBI) trains a neural estimator on forward simulations and applies it to observations [Cranmer et al., 2020, Greenberg et al., 2019, Boelts et al., 2025]. Its validity therefore depends not only on the simulator's physics, but also on whether the training simulations represent the inputs encountered during deployment.

Short clips are the practical training unit: amortized training needs a large corpus of simulated videos, and clip length sets the cost of simulating, rendering, and training on each one. Experimental recordings are longer because the temporal evolution they contain is itself of interest, so a recording is analyzed through consecutive windows of training length. In the audit, the continuous simulation starts from the training initialization distribution, and later windows inherit changes in molecular populations, spatial correlations, and photophysical state from the preceding dynamics (Fig. 1). We call this difference the *horizon mismatch* and the assumption that it does not affect inference the *reset assumption*. It is one form of deployment mismatch: the states and histories presented to the estimator differ from those represented in training, even though the simulator itself is unchanged.

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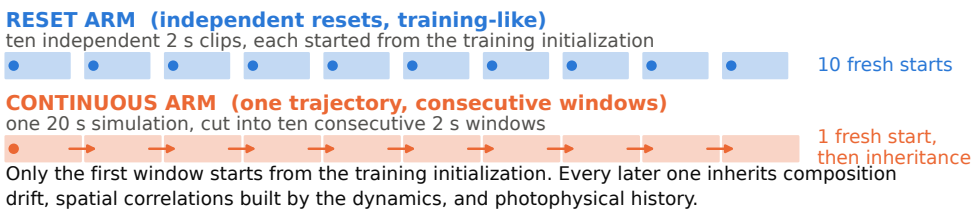


Figure 1: **Testing the reset assumption.** Training uses independently initialized 2 s clips. In the continuous audit simulation, one 20 s trajectory is divided into ten consecutive windows, of which only the first begins from the training initialization distribution. (A_0, B_0, C_0) are the initial counts of free monomeric receptors, mobile dimers, and immobilized dimers. The composition truth for each window is its exact first-frame population; a matched reset clip starts again from (A_0, B_0, C_0).

Both audit arms share the same parameters, dynamics, imaging model, and prior, and every window or clip is a 2 s input to the same estimator; the arms differ only in whether each window is initialized independently or inherits the history of a continuous trajectory.

That finite-time statistics depend on initialization and history is well established for stochastic processes, including aging and non-ergodic systems [Metzler et al., 2014], but this does not determine how a trained neural estimator responds to such history. Studies of SBI robustness have examined perturbed observations, discrepancy models, incompatible summaries, and test-time distribution shifts [Cannon et al., 2022, Ward et al., 2022, Frazier et al., 2024, Schmitt et al., 2023], and calibration diagnostics assess an estimator relative to a specified simulator distribution [Talts et al., 2018, Lemos et al., 2023]; if held-out simulations repeat the reset initialization used for training, they cannot reveal errors caused by inherited state.

Compositional SBI for time series addresses long sequences by learning single-state transitions from a proposal intended to cover the reachable state space [Gloeckler et al., 2025]. We ask the complementary question: how accurately does an already-trained short-window estimator operate when applied unchanged along a continuous trajectory? In single-molecule applications, simulator mismatch is recognized as an important limitation [Dingeldein et al., 2025], but existing benchmarks evaluate independently simulated trajectories or fixed-duration fields of view [Muñoz-Gil et al., 2021, 2025]. DeepSPT combines a simulation-trained segmentation module with classifiers trained on annotated experimental data [Kæstel-Hansen et al., 2025]. Among these studies, none reports a controlled reset-versus-continuous comparison with known window-specific truth.

We contribute a paired reset-versus-continuous audit with exact simulator truth and prespecified operational tolerances [Lakens et al., 2018]; evidence that horizon mismatch affects composition, kinetics, and diffusion differently within one single-particle-tracking inference workflow; and the use of the measured horizon-mismatch error as a sensitivity test, not a correction, for the workflow's conclusions from experimental MET recordings.

2 System: simulator, estimator, and deployment

Simulator. We represent membrane-receptor dimerization as the particle-based reaction–diffusion process $A + A \rightleftharpoons B \rightleftharpoons C$, where A is a freely diffusing monomeric receptor, B a mobile dimer, and C an immobilized dimer represented as an effective state. Simulations use ReaDDy [Hoffmann et al., 2019] and follow earlier amortized-inference work on spatial-stochastic biological systems [Ramírez Sierra and Sokolowski, 2025]. The ten parameters have log-uniform priors and comprise the initial counts of A, B, and C; the monomer diffusion coefficient D_{A} ($\mu\mathrm{m}^2\,\mathrm{s}^{-1}$); two relative-mobility ratios; an association-efficiency parameter R_{ON}, defined relative to the diffusion-limited association ceiling; and the rates of dimer unbinding (κ_{OFF}), immobilization, and remobilization (s^{-1}). Particle trajectories are rendered as synthetic videos using a fixed, experiment-informed microscopy model.

Estimator. Each 2 s input contains 100 video frames. A five-block three-dimensional convolutional network extracts local spatiotemporal features, and a two-layer temporal Transformer inte-

grates them across the clip into a 128-dimensional representation [Vaswani et al., 2017]. This representation conditions a masked autoregressive flow that approximates the joint posterior over the ten model parameters [Papamakarios et al., 2017]. The estimator is trained by neural posterior estimation [Greenberg et al., 2019] using the `sbi` toolkit [Boelts et al., 2025].

For each window we draw 1,000 joint posterior samples; directly inferred parameters are summarized by their marginal posterior medians. The *dimer-complex fraction*, $f_{\mathrm{D}} = (B+C)/(A+B+C)$, the fraction of simulated molecular complexes that are dimers, is calculated within every joint draw, preserving dependence among the three counts, and summarized by its posterior mean and central 90% interval. On 10,000 independently generated held-out videos, the mean absolute error is 2.8 percentage points for f_{D}. For D_{A} and κ_{OFF}, the log-scale mean absolute errors correspond to multiplicative factors of $\times 1.05$ and $\times 1.50$, respectively. The association-efficiency parameter R_{ON} is not identifiable from the tested 2 s videos, so no subsequent scientific claim is based on it. Training and data-split details are given in Appendix G.

Experimental deployment. MET is a receptor tyrosine kinase of the plasma membrane; binding of its ligands, including the bacterial protein InlB, is associated with receptor dimerization and signaling. Single-particle tracking follows individually labeled receptors in living cells. We apply the estimator directly to MET recordings acquired for single-particle tracking [Harwardt et al., 2017]: 25 recordings of resting MET labeled with a non-activating Fab fragment and 25 recordings of MET labeled and activated by fluorescent InlB. Independent single-molecule studies report increased MET dimerization after ligand binding [Dietz et al., 2013, Baldering et al., 2021], which supports describing the InlB condition as dimerization-enriched but provides no stoichiometric ground truth for these recordings.

Each 20 s recording is divided into ten consecutive 2 s windows, matching the estimator's training duration. The estimator produces one result per window; the ten results are averaged to one value per cell, the independent biological replicate, so horizon mismatch is evaluated at both the window and the recording level.

3 The horizon audit

Design. For every parameter setting, we generate two matched audit arms with identical parameters and imaging. The *continuous* arm is one 20 s simulation divided into ten consecutive 2 s windows. The *reset* arm contains ten independently initialized 2 s simulations. Every window and every clip is a 2 s input to the fixed estimator used for the experimental recordings, so the arms differ in initialization and inherited history, not in input length. Two of 502 attempted settings were rejected, leaving 500 (disposition, seeding, and cohort tracking in Appendix F).

Truth reference. The simulator stores the exact A, B, and C populations at every frame of the rendered trajectory, so the composition truth belongs to the same stochastic realization as the estimator input and is not reconstructed from the image. Training labels represent the population at the beginning of a clip, so the primary truth for window j is its exact first-frame population, from which f_{D} is calculated. At the recording level, we average both the ten inferred values and the ten window-start truths. Alternative truth definitions are examined in Appendix F.

Statistics. Each matched parameter setting is one independent statistical unit; the ten windows from its continuous trajectory are correlated repeated measurements [Lazic, 2010]. We calculate 95% percentile confidence intervals from 2,000 bootstrap resamples of the 500 settings. For setting i and window j, the primary statistic is the difference between the continuous-window absolute error and the mean absolute error of the ten matched reset clips:

$$\Delta_{ij} = \left|\hat{y}_{ij}^{\mathrm{cont}} - y_{ij}^{\star}\right| - \frac{1}{R}\sum_{r=1}^{R}\left|\hat{y}_{ir}^{\mathrm{reset}} - y_{ir}^{\star}\right|, \qquad R = 10. \tag{1}$$

Positive values indicate lower accuracy in the continuous arm. For constant parameters, $y_{ij}^{\star} = \theta_i^{\star}$; for f_{D}, the truth changes with the window-start population. The analogous contrast without absolute values is called the *signed shift* and describes the direction of the inference error.

The three primary quantities were declared before the audit: the dimer-complex fraction f_{D}, dimer unbinding rate κ_{OFF}, and monomer diffusion coefficient D_{A}. Individual species counts and the

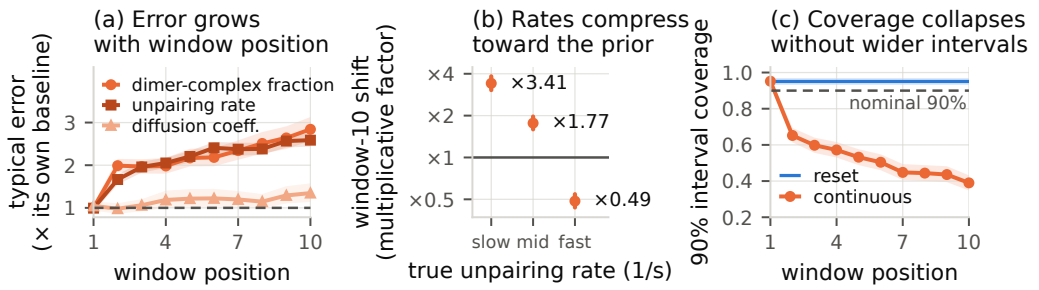


Figure 2: **Horizon mismatch affects the three quantities differently.** (a) Mean absolute error by window position, divided by each quantity's held-out error (dashed line). Composition and unbinding-rate errors approach three times their baselines, whereas diffusion changes modestly. Verdicts use the matched horizon contrast in Table 1. (b) Signed unbinding-rate shift at window 10 by true-rate third: slow rates are read high and fast rates low, indicating compression toward the prior interior. (c) Coverage of the central 90% interval for f_{D}. Coverage in the continuous arm falls to 0.39 without wider intervals. Bands and bars are 95% bootstrap intervals over matched settings.

Table 1: Results over 500 matched parameter settings. Total error is the mean absolute error; the horizon contrast is continuous minus reset absolute error, expressed for the log-scale parameters as a ratio of error factors ($\times 1$: no additional error). Verdicts compare the entire 95% confidence interval with the tolerance; recording-level values average ten windows under the same tolerances.

quantity, level	total error (reset → cont.)	horizon contrast [95% CI]	tolerance	verdict
f_{D}, window 10	2.9 → 8.3 pp	+5.4 [4.7, 6.2] pp	2.8 pp	degraded
f_{D}, recording	1.7 → 5.6 pp	+3.9 [3.4, 4.4] pp	2.8 pp	degraded
κ_{OFF}, window 10	×1.53 → ×3.02	×1.97 [1.85, 2.10]	×1.50	degraded
κ_{OFF}, recording	×1.38 → ×2.22	×1.61 [1.55, 1.68]	×1.50	degraded
D_{A}, window 10	×1.05 → ×1.07	×1.02 [1.01, 1.03]	×1.05	equivalent
D_{A}, recording	×1.03 → ×1.05	×1.01 [1.01, 1.02]	×1.05	equivalent

remaining parameters are treated as exploratory diagnostics. Each primary quantity has a prespecified tolerance equal to its mean absolute error on independently initialized held-out videos. This estimator-relative operational tolerance, not a biological effect-size threshold, asks whether horizon mismatch adds at least as much error as the estimator already carries under its validation conditions.

Following equivalence-testing logic [Lakens et al., 2018], a quantity is *degraded* when its entire confidence interval lies above the positive tolerance, *equivalent* when the interval lies within the negative and positive tolerance bounds, and *inconclusive* otherwise. Comparison with zero alone establishes detectability, not whether the additional error is large relative to baseline performance. Before interpreting later windows, the prespecified first-window gate requires all three primary quantities to be equivalent within their respective tolerances.

4 Results

4.1 Horizon mismatch affects the primary quantities selectively

At window 1, every continuous-minus-reset confidence interval lies within its prespecified tolerance: +0.00 pp [−0.22, +0.22] for f_{D}, ×1.00–×1.01 for D_{A}, and ×0.97–×1.03 for κ_{OFF}. The first-window gate therefore passes: the arms are comparable before recording history accumulates.

By window 10, the dimer-complex fraction and unbinding-rate errors exceed their tolerances, whereas the diffusion-coefficient error remains within tolerance (Table 1). Averaging the ten windows reduces total error but leaves all three verdicts unchanged, so the horizon-dependent error persists at the recording level.

For f_{D}, lower accuracy is not accompanied by greater posterior uncertainty: the central 90% intervals cover the truth in 95% of settings at window 1 but only 39% at window 10 while their widths

do not grow (Appendix F), so posterior width alone does not reveal this mismatch. The degradation also persists when populations remain within the training initialization range, so range escape alone does not explain it (Appendix B).

4.2 Unbinding-rate estimates shift toward the training-range interior

Because held-out accuracy varies with the true value, we repeated the analysis in prespecified parameter strata with local tolerances (Appendix A). Composition is degraded in both adequately sampled strata; the highest-composition stratum contains only two settings and is not assessed. The unbinding rate is degraded in the slow and fast thirds and inconclusive in the middle. For D_{A}, the lowest third is equivalent and the remaining two are inconclusive; none is degraded.

The signed κ_{OFF} shift decreases monotonically with the true rate (Fig. 2b). Relative to the matched reset arm, window-10 estimates are shifted by factors of $\times 3.41$, $\times 1.77$, and $\times 0.49$ across the slow, intermediate, and fast thirds. Thus, slow rates are estimated too high and fast rates too low, compressing estimates toward the prior interior. This pattern is consistent with reduced information about the true rate, although the audit does not identify the estimator-level cause (Appendix C). The compression remains after recording-level aggregation.

4.3 Sensitivity of conclusions from experimental MET recordings

For the MET recordings, the estimator reports the relative abundance of simulated complex categories, not molecular stoichiometry. The inferred dimer-complex fraction is $31.4 \pm 2.0\%$ for resting MET-FAB and $78.2 \pm 3.7\%$ for activated MET-INLB (mean $\pm$ s.e.m., $n = 25$ cells per condition). The cell-level distributions differ in a Mann–Whitney U test ($p \approx 2 \times 10^{-9}$). Within MET-INLB recordings, the mean decreases from 87.0% in the first window to 70.2% in the last (paired Wilcoxon test, $p = 0.0015$).

We use the simulated recording-level shifts as an uncalibrated sensitivity scale for the experimental conclusions (transfer rules and absolute values in Appendices D and E), not as corrections or uncertainty bounds.

The condition contrast remains supported under direct transfer. The difference between MET-FAB and MET-INLB changes from $46.8\,\mathrm{pp}$ to about $48.3\,\mathrm{pp}$ (Fig. 4, with the inferred time courses); the simulated horizon effect is therefore too small to generate the observed condition difference.

The horizon effect does not generate the within-recording decline. Inherited history shifts late-window f_{D} upward in simulation, opposite to the observed $87.0\% \rightarrow 70.2\%$ decrease; it would therefore mask part of the decline rather than create it.

Absolute composition and unbinding-rate estimates remain conditional. The transfers establish relative scale only; they neither correct the experimental estimates nor identify the cause of their temporal trends.

5 Discussion and conclusions

Reliability is quantity-specific. Because the arms pass the first-window gate, later degradation is caused by what later windows inherit, not by clip duration alone. Under the tested mismatch, D_{A} remains within tolerance, whereas f_{D} loses accuracy and interval coverage, and averaging does not remove the additional error. The unbinding-rate error is larger and rate-dependent, with slow rates shifted upward and fast rates downward, so a whole-recording estimate cannot be interpreted as one constant unbinding rate. For the MET recordings, direct transfer is too small to explain the condition contrast, although absolute fractions remain conditional.

Scope and implication. This audit isolates a reset-trained estimator applied to history-inheriting windows within one model; it neither tests other misspecification nor explains experimental trends, distinguishes mechanisms, or provides a correction. Since held-out reset clips cannot establish performance after stateful evolution, validation should reproduce the deployed inference sequence, use the trajectory as the statistical unit, and evaluate each reported quantity and aggregation rule against simulator truth; training can instead use full-length simulations or clips from reachable states. Reliability must be established per conclusion, not for the estimator as a whole.

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5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The release repository (`https://github.com/covinolab/short-simulators-long-experiments`) contains the simulator and estimator code at the version used for this study, with the version and commit recorded; the persisted audit, recovery, and experimental-inference results behind every figure and table; the scripts that regenerate the figures and recompute the supplied numerical checks from those results; and an environment specification. The repository regenerates all manuscript figures and recomputes the supplied numerical checks from persisted results. Recomputing the full audit analysis and the posterior interval-width statistics requires the per-setting files, available from the authors on request; the video corpora and trained estimator are not included. The README states these limits and names the public accession of the experimental recordings. Sections 2–3 and Appendices F–G specify the model, estimator, experimental design, statistics, data splits, training configuration, seeding, and cohort-verification scheme.

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   Answer: [Yes]

Justification: Section 2 describes the simulator, estimator architecture, training horizon, posterior summaries, and held-out recovery errors. Section 3 gives the audit durations, cohort size, replication, and fixed imaging model. Appendix G reports the three independent data splits and their roles, sample counts, objective, optimizer, batch size, epoch count, learning-rate schedule, checkpoint criterion, augmentation, flow settings, and seeding policy.

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7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Every interval is a bootstrap over the 500 matched parameter settings. Windows within one continuous trajectory are correlated repeated measures and are never resampled as independent replicates. Verdicts use three-outcome comparisons against tolerances fixed before the experiment, and stratified tolerances are reported so a prior-averaged yardstick cannot hide a regime-specific effect.

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Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Appendix H reports CPU and GPU worker types, allocated and observed memory, storage, concurrency, measured wall times, and explicitly labelled projections for wall times that were not retained. It separately accounts for corpus generation, estimator training, 10,000-video held-out recovery, and the 500-setting horizon audit, and discloses the additional 275,000-video corpus used by the wider two-stage project.

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    Answer: [Yes]

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A Stratified tolerances

This appendix gives the stratified verdicts summarized in Sec. 4.2. For D_{A} and κ_{OFF}, the strata are prespecified equal thirds of the log-prior range; because the priors are uniform in log space, these thirds have equal expected mass. The f_{D} strata are the prespecified intervals 0–0.2, 0.2–0.8, and 0.8–1 of the true composition at the start of window 10, the window at which the contrast is evaluated; because dimers dissociate along the 20 s trajectory, 417 of the 500 settings lie in the lowest interval at window 10, although 230 start in the highest one (by the true composition in the first frame, with the same intervals). None of the boundaries is chosen from the observed effects. Each stratum is compared with the estimator's held-out recovery error in the same part of the prior. For a log-scale quantity, a displayed tolerance $\times q$ represents the symmetric interval $[\times 1/q, \times q]$, and $\times 1$ denotes no horizon effect. The local f_{D} tolerances are constructed from held-out count medians (prior-average $2.81\,\mathrm{pp}$), whereas audit values are formed within joint posterior draws; the small difference from the declared $2.8\,\mathrm{pp}$ tolerance is retained explicitly.

stratum (true value at window 10)	n	local tolerance	horizon contrast [95% CI]	signed shift (window 10)	verdict
κ_{OFF} 0.10–0.46 s^{-1}	162	$\times 1.622$	$\times 2.938$ $[2.624, 3.296]$	$\times 3.41$	degraded
κ_{OFF} 0.46–2.15 s^{-1}	164	$\times 1.439$	$\times 1.355$ $[1.250, 1.479]$	$\times 1.77$	inconclusive
κ_{OFF} 2.15–10 s^{-1}	174	$\times 1.472$	$\times 1.932$ $[1.758, 2.138]$	$\times 0.49$	degraded
D_{A} 0.056–0.121 $\mu\mathrm{m}^2/\mathrm{s}$	169	$\times 1.057$	$\times 0.991$ $[0.984, 0.998]$	$\times 0.97$	equivalent
D_{A} 0.121–0.261 $\mu\mathrm{m}^2/\mathrm{s}$	170	$\times 1.052$	$\times 1.033$ $[1.016, 1.054]$	$\times 0.96$	inconclusive
D_{A} 0.261–0.562 $\mu\mathrm{m}^2/\mathrm{s}$	161	$\times 1.050$	$\times 1.028$ $[1.009, 1.057]$	$\times 0.97$	inconclusive
f_{D} 0–0.2	417	$\pm 1.85\,\mathrm{pp}$	$+4.62$ $[3.86, 5.43]\,\mathrm{pp}$	$+7.0\,\mathrm{pp}$	degraded
f_{D} 0.2–0.8	81	$\pm 3.60\,\mathrm{pp}$	$+8.81$ $[7.11, 10.54]\,\mathrm{pp}$	$-7.1\,\mathrm{pp}$	degraded
f_{D} 0.8–1	2	$\pm 2.34\,\mathrm{pp}$	$+28.80$ $[26.31, 31.28]\,\mathrm{pp}$	$-30.2\,\mathrm{pp}$	not assessed

The signed shifts show the compression directly: slow and intermediate κ_{OFF} values are read high, by factors of 3.41 and 1.77, whereas fast values are read low, by a factor of 0.49. The middle stratum remains inconclusive because its horizon-contrast confidence interval crosses the local tolerance, so its directional shift is not itself a degradation verdict. No verdict is assigned to the highest-composition stratum because it contains only two audit settings.

For scale, a true $\kappa_{\mathrm{OFF}} = 0.2\,\mathrm{s}^{-1}$ corresponds to a characteristic unbinding timescale of $5\,\mathrm{s}$. Applying the $\times 3.41$ mean upward shift observed in the slowest stratum would give approximately $0.68\,\mathrm{s}^{-1}$, corresponding to about $1.5\,\mathrm{s}$. The quantity $1/\kappa_{\mathrm{OFF}}$ is a characteristic unbinding timescale, not the full complex lifetime, because this reaction scheme also includes transitions through the immobilized state C.

The mean D_{A} horizon contrast is smaller than its local tolerance in every stratum. The lowest stratum is equivalent; the other two are inconclusive because their confidence intervals cross the tolerance. None is degraded.

B State-support escape

This appendix quantifies how far the continuous windows leave the initialization range used for training, and whether leaving it is needed for the degradation. Across the 5,000 continuous windows, 52.5% have a true starting population outside the count box from which training clips were initialized. Escape occurs at both ends for different reasons. Species B is absent in 27% of windows and C in 25%: extinction is dynamically reachable, but a log-uniform count prior cannot represent zero. At the other end, free monomers exceed their initialization ceiling in 20% of windows because dissociation converts each dimer into two monomers. Total receptor number is conserved exactly, so these changes reflect repartitioning among states rather than receptor creation.

The practical consequence is that a box prior on initial per-species counts is not closed under the model's dynamics: trajectories initialized inside it can leave it through valid dynamics. Training initializations intended to cover reachable deployment states therefore need to represent extinction and a higher monomer ceiling. This is a statement about state support, not about estimator behavior, and count-support escape is not necessary for the degradation reported in Sec. 4.1. Among the 92 settings that remain inside the box in every window, the added f_{D} error is $+5.5\,\mathrm{pp}$ $[4.0, 6.9]$, close to

the full-cohort value of $+5.4\,\mathrm{pp}$ $[4.7, 6.2]$. The overlapping intervals do not establish equal effects, but both exceed the same prespecified tolerance.

C Candidate explanations for the direction pattern

This appendix records the exploratory analysis behind the direction pattern of Sec. 4.2; it does not identify a mechanism. We first tested whether window position contains information beyond the measured state variables. A within-trajectory regression included starting f_{D}, its signed and absolute change during the window, and the fraction of posterior samples outside the parameter ranges used for training. Adjustment changes the correlation between window position and κ_{OFF} error from $r = +0.105$ to $r = -0.05$, while the covariates explain approximately 11% of the error variance. Window position therefore overlaps with the measured covariates, but this analysis neither establishes mediation nor identifies a causal mechanism.

Two inherited-state channels could contribute, but the present audit ranks them only by their predicted directions. Spatial memory is one candidate: after dissociation, two monomers remain nearby and can rebind sooner than independently positioned monomers. A reset-trained estimator could interpret this rapid local cycling as a larger unbinding rate while also overestimating f_{D}, matching the observed directions. A simple accumulated-photophysics explanation would instead tend to lower apparent f_{D}, so the observed upward shift is less consistent with that explanation alone.

These sign comparisons do not identify a mechanism. The audit does not reset spatial and photophysical history separately. Two controlled ablations would separate them: reset photophysics at each continuous-window boundary while preserving molecular state, or retain the true counts while redistributing particles uniformly to remove spatial correlations. Neither ablation has been run.

D Claim verdicts

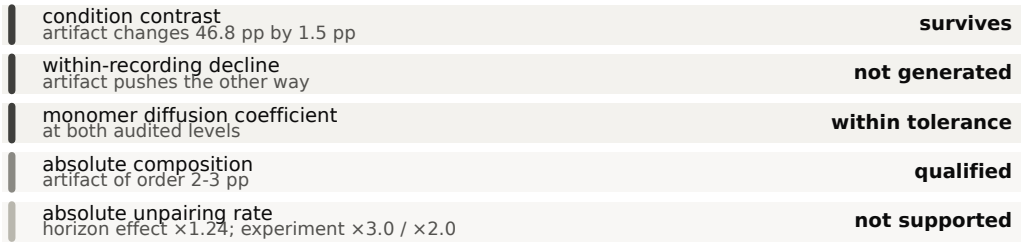


Figure 3: Grading each claim against the measured horizon effect. Because the audit gives the effect's size and sign under the simulator, it permits a claim-specific sensitivity assessment. Verdicts are carried by the words rather than by color, since blue and orange denote the two arms throughout this paper. Simulator-to-experiment transfer is uncalibrated, so the effect serves as a sensitivity scale rather than as a correction to any measurement.

Using the same log-linear summary across window positions, the simulated horizon effect increases the inferred unbinding rate by a factor of approximately $\times 1.24$ across a recording, compared with experimental increases of approximately $\times 3.0$ for MET-FAB and $\times 2.0$ for MET-INLB. This comparison establishes relative scale only; it neither identifies the cause of the experimental trends nor supports interpreting a whole-recording estimate as one constant unbinding rate.

E The census contrast

Background for readers outside receptor biology. MET is a receptor tyrosine kinase of the plasma membrane. Its activation by hepatocyte growth factor, or by the *Listeria monocytogenes* surface protein InlB, is associated with receptor dimerization and downstream signaling [Dietz et al., 2013, Baldering et al., 2021]. In single-particle tracking, individual receptors carry a fluorescent label and are imaged at video rate in living cells; fluorescent spots are detected and linked across

frames to reconstruct trajectories and estimate receptor mobility [Harwardt et al., 2017]. The recordings used here label MET either with a non-activating Fab fragment (resting condition) or with fluorescent InlB (activated condition), 25 cells each. The estimator reads every video as a realization of the simulated $A + A \rightleftharpoons B \rightleftharpoons C$ model, so the inferred composition is a model-conditional quantity, not a stoichiometric count.

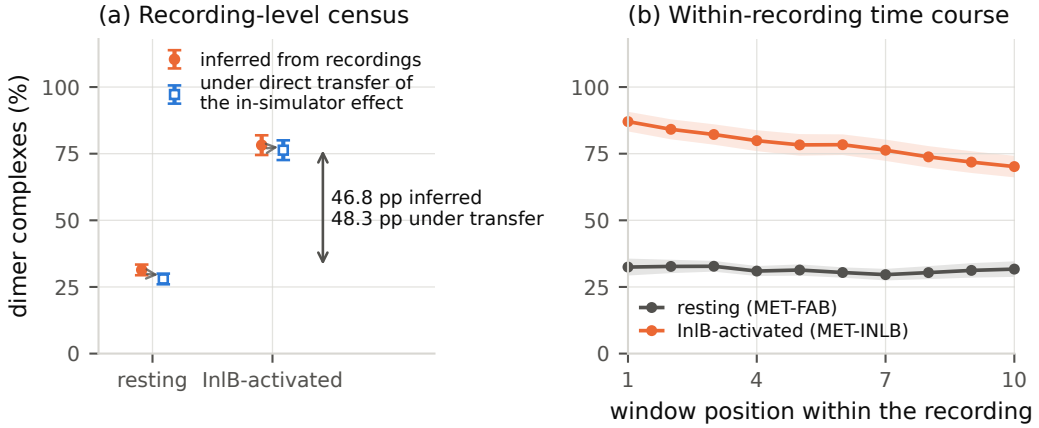


Figure 4: Inference results from experimental MET recordings (Sec. 4.3). (a) The model-conditional census of dimer complexes on MET single-particle-tracking video, as inferred and under direct transfer of the in-simulator horizon effect. The transferred effect lowers both conditions, the resting one slightly more, so the contrast changes by only $1.5\,\mathrm{pp}$. This is a sensitivity calculation: simulator-to-experiment transfer is uncalibrated, so these are not corrected experimental estimates. Direct-transfer values subtract the recording-level signed shift of simulated settings with initial $f_\mathrm{D} < 0.4$ (resting-like, $n = 115$) or $f_\mathrm{D} > 0.7$ (activated-like, $n = 276$). (b) The inferred dimer-complex fraction by window position, averaged over the 25 cells of each condition (band: cell-level s.e.m.); the activated condition declines from 87.0% to 70.2% while the resting condition remains near 31%. Only experimental inferences are drawn; in simulation, inherited history shifts late windows upward ($+3.5\,\mathrm{pp}$ at window 10 for activated-like settings), so it does not generate this decline. All error bars and bands are cell-level standard errors; they do not represent uncertainty in the transfer.

Direct transfer changes the absolute composition estimates from 31.4% and 78.2% to approximately 28% and 76%, respectively. These values illustrate sensitivity to the simulated horizon effect and are not corrected experimental estimates.

The transfer subsets are defined by the initial composition label of each setting ($f_\mathrm{D} < 0.4$: $n = 115$; $f_\mathrm{D} > 0.7$: $n = 276$ of 500 settings), whereas the strata of Appendix A are defined by the true composition at the window-10 start; the two groupings answer different questions. Grouped by initial composition, the signed window-10 shift of f_D is upward in both subsets, $+5.6\,\mathrm{pp}$ $[4.1, 7.2]$ for resting-like and $+3.5\,\mathrm{pp}$ $[2.0, 5.1]$ for activated-like settings, and the recording-level means used for the transfer are $+3.4\,\mathrm{pp}$ $[2.6, 4.3]$ and $+1.9\,\mathrm{pp}$ $[0.7, 3.0]$ (95% bootstrap intervals over settings). The negative window-10 shifts of the two higher strata in Appendix A describe the smaller group of settings that still hold a mid or high composition at window 10. The statement in Sec. 4.3 that inherited history shifts late-window f_D upward therefore holds for the activated-like subset on which the within-recording comparison relies, and the transfer remains a sensitivity calculation for that selected scenario rather than a bound on experimental bias.

F Truth reference and reproducibility

This appendix documents how the cohort was completed and seeded, and why the window-start population is the truth reference for f_D. Sampling continued until 500 settings were completed. Two of 502 attempted settings exceeded the available rendering memory and remain documented as rejected draws. The analyzed cohort is therefore sampled from the training prior conditional on renderability.

A persisted master seed deterministically generates one seed for every (setting, arm, replicate) combination. Each result file also carries a cohort digest, which the analysis verifies before use to prevent results from different cohorts being mixed.

The estimator's training labels are window-start populations, so the window start is the primary truth reference for f_{D}. The table reports descriptive mean absolute error across 500 settings and ten clips or windows per setting in each arm. Confidence intervals elsewhere resample matched settings, not the correlated windows. Evaluating the same estimates against the within-window mean asks about time-averaged composition rather than the start-state estimand used for training. Under that alternative definition, the reset-arm error is $27\,\mathrm{pp}$ rather than $2.9\,\mathrm{pp}$; this difference reflects a changed estimand, not a failure of the estimator on its training target.

Interval widths. The coverage decline of Sec. 4.1 is not accompanied by wider intervals. From the saved posterior draws of all 500 settings, the mean width of the central 90% interval for f_{D} is $15.1\,\mathrm{pp}$ $[14.5, 15.8]$ at window 1 and $13.6\,\mathrm{pp}$ $[12.8, 14.4]$ at window 10 in the continuous arm, against $15.2\,\mathrm{pp}$ $[14.5, 15.9]$ for the reset clips (95% bootstrap intervals over settings); the paired change from window 1 to window 10 is $-1.5\,\mathrm{pp}$ $[-2.1, -0.9]$. The central 50% interval narrows from $6.1\,\mathrm{pp}$ to $5.5\,\mathrm{pp}$ over the same windows, with $6.1\,\mathrm{pp}$ for the reset clips. The intervals therefore narrow slightly while coverage falls from 0.95 to 0.39.

truth reference for f_{D}	reset arm	continuous arm (all windows)
window start (primary)	$2.93\,\mathrm{pp}$	$6.33\,\mathrm{pp}$
window mean (sensitivity)	$27.00\,\mathrm{pp}$	$9.00\,\mathrm{pp}$
window end (sensitivity)	n/a	$10.34\,\mathrm{pp}$
unrounded parameter label (sensitivity, resets only)	$2.89\,\mathrm{pp}$	n/a

G Estimator training and data splits

The estimator audited in this study was trained once and then held fixed; we did not retrain or tune it for either audit arm or for the experimental recordings. All videos in these three datasets were independently generated 2 s grayscale clips with 100 frames of 256×256 pixels. Three physically separate datasets had distinct roles: 200,000 TRAIN videos supplied gradient updates, 50,000 TEST videos supplied the per-epoch model-selection loss, and 10,000 EVAL videos were used only for final held-out recovery and for setting the audit tolerances reported in Sec. 2. No video was shared among these datasets. The tolerances were computed from the recovery's optimized point estimates (Appendix H); summarizing the same 10,000 videos by marginal posterior medians gives log-scale mean absolute errors of 0.0222 dex ($\times 1.053$) for D_{A} and 0.1788 dex ($\times 1.509$) for κ_{OFF}, against 0.0227 dex ($\times 1.054$) and 0.1773 dex ($\times 1.504$) from the optimized estimates; using either recovery error as the tolerance leaves the primary audit verdicts unchanged. The f_{D} tolerance of $2.8\,\mathrm{pp}$ is composed from the count medians. Table 2 lists the prior ranges and the fixed simulation and imaging settings behind every video in this study.

Training minimized the mean negative conditional log density of the true log-transformed parameter vector for 50 epochs. We used AdamW with a minibatch of 32 videos per process, an initial learning rate of 1.28×10^{-3}, and the PyTorch 2.9.1 defaults $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, and weight decay 0.01. A `ReduceLROnPlateau` scheduler monitored TEST loss with reduction factor 0.5, patience one epoch, absolute improvement threshold 10^{-4}, and minimum learning rate 10^{-5}. After two non-improving epochs at the floor, the enabled restart rule restored the best checkpoint and reduced the preceding peak learning rate by a factor of four; restarts ended when the next peak would not exceed the floor. The retained checkpoint was the epoch with the lowest TEST loss; its mean loss was -7.4801.

TRAIN videos were augmented on loading by a uniformly sampled multiple of a 90° rotation and independent horizontal and vertical flips, each with probability 0.5; augmentation was disabled for TEST and EVAL. Integer video values were scaled to $[0, 1]$, and parameter targets were transformed to base-10 logarithms. The conditional masked autoregressive flow contained five autoregressive transforms, each with two residual blocks and 50 hidden features. It used structured standardization of parameter targets and conditioning videos, batch normalization, and dropout probability 0.1.

Table 2: Configuration of the simulator, the imaging model, and the prior. Priors are log-uniform between the stated bounds. The six emitter values are the calibrated vector of the training corpus (the sample geometric median of imaging estimates on the resting MET recordings), fixed for every training video and for both audit arms. The five camera values were marginalized in training: each training video drew them from the narrow log-uniform boxes given in brackets, and both audit arms fixed them at the listed reference values, which are the acquisition configuration for gain, baseline, and quantum efficiency, the camera datasheet for read noise, and the median photon offset measured on the MET recordings for the optical background.

group	quantity	value
prior	initial counts A_0, B_0, C_0	1 to 316 each
	monomer diffusion coefficient D_{A}	0.056 to 0.56 $\mu m^2/s$
	relative mobility D_B/D_{A}, D_C/D_{A}	0.24 to 0.75; 0.01 to 0.10
	association efficiency R_{ON}	0.01 to 1
	unbinding, immobilization, remobilization rates	0.1 to 10 s^{-1} each
simulation	field of view	256×256 pixels of 158 nm (40.4 µm square)
	box depth; boundaries	2 nm (periodic); open in the plane
	particle diameter (contact distance)	10 nm
	frame interval; integration step	0.02 s (50 frames/s); 2 ms (10 steps per frame)
	frames per 2 s clip; per 20 s trajectory	100; 1,000
emitter	PSF width, median $\sqrt{2}\sigma$	1.69 pixels (log-normal spread 0.18)
	label brightness, median	251 photons per frame (log-normal spread 0.67)
	brightness flicker rate; bleaching	3.39 s^{-1}; probability 0.107 per 100 frames
	dimer	two independent labels, photons added
camera	gain	41.8 ADU per photoelectron [41.7 to 42.2]
	quantum efficiency	0.90 [0.89 to 0.91]
	optical background	28.7 photons per pixel and frame [28.5 to 29.2]
	baseline; read noise	175 ADU [174 to 178]; 10.5 ADU [10.5 to 10.6]
	storage	8-bit integers, fixed range

TRAIN/TEST/EVAL generation and estimator training did not impose a fixed random seed; the generated datasets and retained estimator were persisted. The code at the version used for this study, the persisted results behind every figure and table, and the scripts that regenerate them are released at `https://github.com/covinolab/short-simulators-long-experiments`; the video corpora are not bundled, and the repository states what can be reproduced from the included materials.

H Computational resources

Computational resources for the reported study and the wider project included the Goethe-NHR cluster at Goethe University Frankfurt, the JUWELS and JUPITER systems at the Jülich Supercomputing Centre, and local workstations. The descriptions below distinguish resources used for the reported study from those used elsewhere in the project. We distinguish measured wall times from scheduler allocations and linear projections. Wall times exclude queueing; the CPU and GPU systems were managed with Slurm.

Synthetic corpus generation. The 2 s biological-state corpus contained 200,000 TRAIN, 50,000 TEST, and 25,000 EVAL videos; 10,000 of the EVAL videos were used for the recovery analysis in this paper. Generation used 28 single-node CPU-array elements (20 TRAIN, five TEST, and three EVAL). Each node ran at most ten independent generation processes with four x86-64 CPU cores and 17.6 GB of requested memory per process, for 40 cores and 176 GB per node. At most 40 nodes were allowed to run concurrently. Each node had an 18 h wall-time allocation, giving an upper allocation budget of 504 node-hours (20,160 core-hours); actual jobs generally ended before that limit. The TRAIN+TEST videos occupy 662.4 GiB.

Estimator training and held-out recovery. A measured timing reference for the same 2 s estimator pipeline used an exclusive node with four AMD Instinct MI210 GPUs (64 GB each), 64 data-loading workers, 480 GB of host memory, and networked scratch storage. With 50,000 TRAIN and 10,000 TEST videos and batch size 32 per process, ten epochs took 13,246 s (3.68 h); steady epochs took approximately 1,267 s. Scaling this measured rate by the number of videos per epoch gives a simple linear reproduction estimate of approximately 74 node-hours, or 296 GPU-hours, for the 200,000/50,000-video, 50-epoch training reported here. This is a workload projection rather than the retained run's wall time. The final held-out recovery was measured directly: 10,000 EVAL videos, sharded evenly over eight MI210 GPUs on a 64-CPU-core, 480 GB node, completed in 4,781 s (1.33 h; 10.6 GPU-hours). Each recovery used 1,000 optimization steps.

Horizon audit. Audit generation ran on a 62 GB x86-64 host. Up to six settings were generated concurrently for typical draws, while dense settings were serialized. A completed setting, comprising one 20 s trajectory and ten independent 2 s resets, took 79–833 s to generate; dense completed settings used approximately 25–35 GB of memory. Two high-reaction-traffic draws exceeded 128 GB and could not be rendered on this host. They remain documented as rejected draws, and the cohort was completed by continuing to sample from the same prior. Processing the 500-setting cohort in batches of 100 limited temporary video storage to approximately 5 GB. Posterior sampling for the resulting 10,000 windows used one NVIDIA RTX 6000 Ada GPU (48 GB) and 1,000 posterior draws per window; pilot timings imply approximately 1.25 GPU-hours for the complete cohort. The latter is a scaled timing estimate because the aggregate inference wall time was not retained.

Compute outside the reported experiment. The wider two-stage project generated a second, separate 275,000-video corpus for imaging-parameter inference, for approximately 550,000 synthetic videos across both stages. It also used four-GPU NVIDIA GH200 nodes for other GPU-heavy analyses. Those costs are not included in the audit totals above.